

Today: Quiz

Review

Worksheet time

Review: 6.1 The Definite Integral

6.2 Fundamental Theorem of Calculus

7.1 Substitution Rule

7.2 Integration by Parts

7.3 Rational Functions and Partial Fractions

7.4 Improper Integrals

§6.1 The Definite Integral

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k) \Delta x$$

$$\Delta x = \frac{b-a}{n}, \quad x_k = a + k \Delta x$$

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

$$\int \frac{1}{x^2+a^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \sec^2 x dx = \tan x + C$$

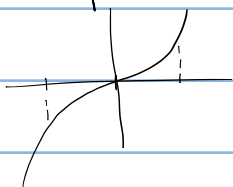
$$\int \cos x dx = \sin x + C$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

Riemann Sum $\sum_{k=1}^n f(x_k) \Delta x$

Example: Approximate $\int_{-1}^1 x^3 dx$ using 2 intervals and right endpoint rule.



$$\Delta x = \frac{1 - (-1)}{2} = 1$$

$$f(x_1) = (0)^3$$

$$0 + 1 = 1$$

$$x_1 = 0, x_2 = 1$$

$$f(x_2) = (1)^3$$

§6.2 The Fundamental Theorem of Calculus

Let $F(x)$ be the antiderivative of $f(x)$.

Then

$$\frac{d}{dx} \int_a^c f(t) dt = f(x)$$

and

$$\int_a^b f(x) dx = F(b) - F(a)$$

Leibniz's Rule

$$\frac{d}{dx} \int_{g(x)}^{h(x)} f(t) dt = f(h(x))h'(x) - f(g(x))g'(x)$$

Example: $\frac{d}{dx} \int_{x^2}^{x^2} \ln(t-3) dt, x > 0$

$$\ln(x^3-3) \cdot 3x^2 - \ln(x^2-3) \cdot 2x$$

§7.1 The Substitution Rule

$$\int f'(g(x))g'(x) dx = f(g(x)) + C$$

Example: $\int (3x^2+6x) \ln(x^3+3x^2) dx$

$$u = x^3 + 3x^2$$

$$du = 3x^2 + 6x$$

$$\int \ln u \, du$$

$$= u \ln u - u + C$$

$$= (x^3+3x^2) \ln|x^3+3x^2| - (x^3+3x^2) + C$$

§7.2 Integration by Parts

$$\int u dv = uv - \int v du$$

Log
Inverse
Alg
Trig
Exp

↑ u
↓ dv

Example: $\int x^2 \cos x dx$

	u	dv
+	x^2	$\cos x$
-	$2x$	$\sin x$
+	2	$-\cos x$
		$-\sin x$

$$x^2 \sin x + 2x \cos x - 2 \sin x + C$$

§7.3 Rational Functions and Partial Fractions

$$f(x) = \frac{P(x)}{Q(x)} \quad P(x), Q(x) \text{ are polynomials}$$

Check if degree of $P(x) >$ degree of $Q(x)$

if so, use long division

Then factor if possible to use partial fractions

Remember how to complete the square to use

$$\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$$

Ex: $\int \frac{1}{x(x+1)^2} dx$

$$\left[\frac{1}{x(x+1)^2} = \frac{A}{x} + \frac{B}{(x+1)^2} + \frac{C}{x+1} \right] x(x+1)^2$$

$$1 = A(x+1)^2 + Bx + Cx(x+1)$$

$$0x^2 + 0x + 1 = Ax^2 + 2Ax + A + Bx + Cx^2 + Cx$$

$$A + C = 0$$

$$A = 1$$

$$2A + B + C = 0$$

§7.4 Improper Integrals

$$\int_0^{\infty} f(x) dx = \lim_{a \rightarrow \infty} \int_0^a f(x) dx$$

$$\int_0^1 f(x) dx = \lim_{a \rightarrow 0^+} \int_a^1 f(x) dx$$

$$\text{If } \lim_{a \rightarrow 0^+} f(x) = \infty$$